

$$1) \lim_{x \rightarrow 1} f(x) = 1 \text{ if } f(x) = \begin{cases} x^2 & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$$

$$\forall \epsilon > 0, \exists \delta > 0 : 0 < |x-1| < \delta \Rightarrow |x^2-1| < \epsilon$$

$$|x^2-1| < \epsilon$$

$$|(x-1)(x+1)| < \epsilon$$

$$|x-1||x+1| < \epsilon$$

$$\text{choose } \delta \leq 1, |x-1| < 1$$

$$-1 < x-1 < 1$$

$$0 < x < 2$$

$$1 < x+1 < 3$$

$$|x-1||x+1| < \epsilon$$

$$|x-1| \cdot 3 < \epsilon$$

$$|x-1| < \frac{\epsilon}{3}$$

$$\text{Then, } \lim_{x \rightarrow 1} f(x) = 1 \text{ iff } \delta = \min\left(\frac{\epsilon}{3}, 1\right)$$

$$b) |f(x)| \leq g(x)$$

$$-g(x) \leq f(x) \leq g(x)$$

$$\text{If } \lim_{x \rightarrow a} g(x) = 0, \text{ by squeeze theorem, } \lim_{x \rightarrow a} f(x) = 0.$$

$$\text{If } \lim_{x \rightarrow a} g(x) = 3, \text{ there is no conclusion, because squeeze theorem can not apply directly.}$$

$$c) f \text{ is continuous at } L$$

$$\forall \epsilon > 0, \exists \delta_1 > 0 : |y-L| < \delta_1 \Rightarrow |f(y)-f(L)| < \epsilon$$

$$\lim_{x \rightarrow c} g(x) = L, \forall \epsilon > 0, \exists \delta_2 > 0 : 0 < |x-c| < \delta_2 \Rightarrow |g(x)-L| < \epsilon$$

$$\text{choose } \delta = \delta_2$$

$$\text{Then, if } 0 < |x-c| < \delta \Rightarrow |f(g(x))-f(L)| < \epsilon$$

$$\text{Thus, } \lim_{x \rightarrow c} f(g(x)) = f(L)$$

$$\textcircled{2} \quad f(x) = \frac{x^2 - 3x + 2}{x^3 - 2x}$$

sign table

x	$-\infty$	$-\sqrt{2}$	0	1	$\sqrt{2}$	2	
$x^2 - 3x + 2$		+	+	0	-	0	+
$x^3 - 2x$	-	0	+	0	-	0	+
$f(x)$	-		+		-	0	+

$$a) \lim_{x \rightarrow 0^+} \frac{x^2 - 3x + 2}{x^3 - 2x} = \frac{0^2 - 3(0) + 2}{0^3 - 2(0)} = \frac{2}{0^+} = +\infty$$

$$b) \lim_{x \rightarrow 2^-} \frac{x^2 - 3x + 2}{x^3 - 2x} = \frac{2^2 - 3(2) + 2}{2^3 - 2(2)} = \frac{0}{4} = 0$$

$$c) \lim_{x \rightarrow 0^-} \frac{x^2 - 3x + 2}{x^3 - 2x} = \frac{0^2 - 3(0) + 2}{0^3 - 2(0)} = \frac{2}{0^-} = +\infty$$

Hence $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$, then $\lim_{x \rightarrow 0} f(x)$ doesn't exist

$$\textcircled{3} \quad 2x + y - \sqrt{x} \sin(xy) = \frac{\pi}{4}$$

First derivative

$$[2x + y - \sqrt{x} \sin(xy)]' = \left(\frac{\pi}{4}\right)'$$

$$2 + y' - \left[\frac{1}{2\sqrt{x}} \sin(xy) + \sqrt{x} (xy)' \cos(xy) \right] = 0$$

$$2 + y' - \frac{1}{2\sqrt{x}} \sin(xy) - \sqrt{x} (y + xy') \cos(xy) = 0$$

$$2 + y' - \frac{1}{2\sqrt{x}} \sin(xy) - \sqrt{x} y \cos(xy) - \sqrt{x} y' \cos(xy) = 0$$

$$y' (1 - x \sqrt{x} \cos(xy)) = \frac{1}{2\sqrt{x}} \sin(xy) + \sqrt{x} y \cos(xy) - 2$$

$$y' = \frac{\frac{1}{2\sqrt{x}} \sin(xy) + \sqrt{x} y \cos(xy) - 2}{1 - x \sqrt{x} \cos(xy)}$$

ntd. ③ $T \equiv y - y_0 = f'(x_0)(x - x_0)$

$$x_0 = \frac{\pi}{4}$$

$$f'(x_0) = \frac{-2 + \frac{1}{2\sqrt{\frac{\pi}{4}}} \sin\left(\frac{\pi}{4}y\right) + \sqrt{\frac{\pi}{4}} \cos\left(\frac{\pi}{4}y\right)y}{1 - \sqrt{\frac{\pi}{4}} \frac{\pi}{4} \cos\left(\frac{\pi}{4}y\right)}$$

$$\frac{1}{2\sqrt{\frac{\pi}{4}}} = \frac{1}{\sqrt{\pi}}$$

$$T \equiv y - y_0 = \frac{\frac{\sin\left(\frac{\pi}{4}y\right)}{\sqrt{\pi}} + \frac{\sqrt{\pi}}{2}y \cos\left(\frac{\pi}{4}y\right)}{1 - \frac{\pi\sqrt{\pi}}{8} \cos\left(\frac{\pi}{4}y\right)} \left(x - \frac{\pi}{4}\right)$$

To find y_0 ,
at $x_0 = \frac{\pi}{4}$

$$2\left(\frac{\pi}{4}\right) + y - \sqrt{\frac{\pi}{4}} \sin\left(\frac{\pi}{4}y\right) = \frac{\pi}{4}$$

$$y - \sqrt{\frac{\pi}{4}} \sin\left(\frac{\pi}{4}y\right) = \frac{\pi}{4} - \frac{\pi}{2}$$

$$y_0 = \sqrt{\frac{\pi}{4}} \sin\left(\frac{\pi}{4}y_0\right) - \frac{\pi}{4}, \text{ then it is difficult to find } y_0.$$

④ $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} + \sqrt{1-x} - 2}{x^2} = \frac{\sqrt{1+0} + \sqrt{1-0} - 2}{0^2} = \frac{0}{0} \text{ (I.F.)}$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x} + \sqrt{1-x} - 2)'}{(x^2)'} = \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{1+x}} - \frac{1}{2\sqrt{1-x}}}{2x} = \frac{0}{0} \text{ (I.F.)}$$

$$\lim_{x \rightarrow 0} \frac{\frac{\sqrt{1-x} - \sqrt{1+x}}{2\sqrt{1+x}\sqrt{1-x}}}{2x} = \lim_{x \rightarrow 0} \frac{\sqrt{1-x} - \sqrt{1+x}}{4x\sqrt{1-x^2}} = \frac{0}{0} \text{ (I.F.)}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1-x} - \sqrt{1+x})(\sqrt{1-x} + \sqrt{1+x})}{4x\sqrt{1-x^2}(\sqrt{1-x} + \sqrt{1+x})} = \lim_{x \rightarrow 0} \frac{(\sqrt{1-x})^2 - (\sqrt{1+x})^2}{4x\sqrt{1-x^2}(\sqrt{1-x} + \sqrt{1+x})}$$

$$(\sqrt{1-x})^2 - (\sqrt{1+x})^2 = 1-x-1-x = -2x$$

$$\lim_{x \rightarrow 0} \frac{-2x}{4x\sqrt{1-x^2}(\sqrt{1-x} + \sqrt{1+x})} = \frac{-2}{4\sqrt{1-0^2}(\sqrt{1-0} + \sqrt{1+0})} = \frac{-2}{8} = -\frac{1}{4}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} + \sqrt{1-x} - 2}{x^2} = \boxed{-\frac{1}{4}}$$

$$\textcircled{5} \lim_{x \rightarrow 3} (x+2) \frac{\sqrt{x^2-6x+9}}{x^2-7x+12} = (3+2) \frac{\sqrt{3^2-6(3)+9}}{3^2-7(3)+12} = \frac{0}{0} \text{ (I.F.)}$$

$$\lim_{x \rightarrow 3} \frac{\sqrt{x^2-6x+9}}{x^2-7x+12} = \frac{\sqrt{(x-3)^2}}{(x-3)(x-4)}$$

$$\sqrt{(x-3)^2} = |x-3|$$

$$|x-3| = \begin{cases} x-3 & \text{if } x \geq 3 \\ -(x-3) & \text{if } x < 3 \end{cases}$$

$$\lim_{x \rightarrow 3} (x+2) \frac{\sqrt{x^2-6x+9}}{x^2-7x+12} = \lim_{x \rightarrow 3} (x+2) \frac{|x-3|}{(x-3)(x-4)}$$

$$\lim_{x \rightarrow 3^+} (x+2) \frac{\cancel{(x-3)}}{(\cancel{x-3})(x-4)} = \frac{3+2}{3-4} = \underline{\underline{-5}}$$

$$\lim_{x \rightarrow 3^-} (x+2) \frac{[-(\cancel{x-3})]}{(\cancel{x-3})(x-4)} = -\frac{(3+2)}{3-4} = \underline{\underline{5}}$$

Hence $\lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x)$, the

$\lim_{x \rightarrow 3} f(x)$ doesn't exist.

$$\textcircled{6} y = \frac{x-1}{x+1}$$

$$y' = \frac{(x-1)'(x+1) - (x-1)(x+1)'}{(x+1)^2}$$

$$= \frac{x+1 - x+1}{(x+1)^2} = \frac{2}{(x+1)^2}$$

$$f'(x_0) = \frac{2}{(x_0+1)^2}$$

$$T \equiv y - y_0 = f'(x_0)(x - x_0)$$

$$y - y_0 = \frac{2}{(x_0+1)^2}(x - x_0)$$

Tangent passes through point $(-1, 0)$

$$0 - y_0 = \frac{2}{(x_0+1)^2}(-1 - x_0)$$

$$\neq y_0 = \neq \frac{2(x_0+1)}{(x_0+1)^2}$$

$$y_0 = \frac{2}{x_0+1}$$

$$\text{Let } y_0 = \frac{x_0-1}{x_0+1} \text{ from our curve}$$

$$f'(x_0) = \frac{2}{(x_0+1)^2} = \frac{2}{(3+1)^2} = \frac{2}{16}$$

$$y_0 = y_0$$

$$\frac{2}{x_0+1} = \frac{x_0-1}{x_0+1}$$

$$y_0 = \frac{2}{x_0+1} = \frac{2}{3+1} = \frac{2}{4}$$

$$x_0 = 2+1 = \underline{\underline{3}}$$

$$y_0 = \underline{\underline{\frac{1}{2}}}$$

$$T \equiv y - y_0 = f'(x_0)(x - x_0)$$

$$y - \frac{1}{2} = \frac{2}{16}(x - 3)$$

$$y - \frac{1}{2} = \frac{2}{8} - \frac{3}{8}$$

$$y = \frac{x}{8} - \frac{3}{8} + \frac{1}{2}$$

$$y = \frac{x}{8} + \frac{-3+4}{8}$$

$$\boxed{T \equiv y = \frac{x+1}{8}}$$

$$f) x^3y - 3xy + x^2y^3 = x$$

$$(x^3y - 3xy + x^2y^3)' = x'$$

$$(x^3y)' - 3(xy)' + (x^2y^3)' = 1$$

$$3x^2y + x^3y' - 3y - 3xy' + 2xy^3 + 3x^2y^2y' = 1$$

$$y'(x^3 - 3x + 3x^2y^2) = 1 - 3x^2y + 3y - 2xy^3$$

$$[y'(x^3 - 3x + 3x^2y^2)]' = (1 - 3x^2y + 3y - 2xy^3)'$$

$$y''(x^3 - 3x + 3x^2y^2) + y'((x^3)' - 3x' + 3(x^2y^2)') = -3(x^2y)' + 3y' - 2(xy^3)'$$

$$(x^3)' = 3x^2$$

$$(x^2y^2)' = 2xy^2 + 2x^2yy'$$

$$(x^2y)' = 2xy + x^2y'$$

$$(xy^3)' = y^3 + 3xy^2y'$$

$$y''(x^3 - 3x + 3x^2y^2) + y'(3x^2 - 3 + 6xy^2 + 6x^2yy') = -6xy - 3x^2y' + 3y' - 2y^3 - 6xy^2y'$$

$$y''(x^3 - 3x + 3x^2y^2) = -3x^2y' + 3y' - 6xy^2y' - 6x^2yy'y' - 6xy - 3x^2y' + 3y' - 2y^3 - 6xy^2y'$$

$$y''(x^3 - 3x + 3x^2y^2) = -6x^2y' + 6y' - 12xy^2y' - 6x^2yy'y' - 6xy - 2y^3$$

$$y''(x^3 - 3x + 3x^2y^2) = -6y'(x^2 - 1 + 2xy^2 + x^2y) - 2y(3x + y^2)$$

$$y'' = \frac{-6y'(x^2 - 1 + 2xy^2 + x^2y) - 2y(3x + y^2)}{x^3 - 3x + 3x^2y^2}$$

$$y'' = \frac{6y'(x^2 - 1 + 2xy^2 + x^2y) + 2y(3x + y^2)}{3x - x^3 - 3x^2y^2}$$

$$(8) i = e^{3t} \sin t$$

$$i = \frac{dq}{dt} \quad (\Rightarrow) \int dq = \int i dt$$

$$q = \int i dt$$

$$q = \int e^{3t} \sin t dt$$

$$u = e^{3t} \quad dv = \sin t dt$$

$$du = 3e^{3t} dt \quad v = -\cos t$$

$$q = -e^{3t} \cos t + 3 \int e^{3t} \cos t dt$$

$$q_1 = \int e^{3t} \cos t dt$$

$$u = e^{3t} \quad dv = \cos t dt$$

$$du = 3e^{3t} dt \quad v = \sin t$$

$$q_1 = e^{3t} \sin t - 3 \int e^{3t} \sin t dt$$

$$= e^{3t} \sin t - 3q$$

$$q = -e^{3t} \cos t + 3(e^{3t} \sin t - 3q)$$

$$q = -e^{3t} \cos t + 3e^{3t} \sin t - 9q$$

$$10q = 3e^{3t} \sin t - e^{3t} \cos t$$

$$q = \frac{1}{10} e^{3t} (3 \sin t - \cos t) + C$$

$$q = \int_0^t i dt = \left[\frac{e^{3t}}{10} (3 \sin t - \cos t) \right]_0^t$$

$$q = \frac{e^{3t}}{10} (3 \sin t - \cos t) - \frac{e^{3 \times 0}}{10} (3 \sin 0 - \cos 0)$$

$$= \frac{e^{3t}}{10} (3 \sin t - \cos t) + \frac{1}{10}$$

$$q = \frac{1 + e^{3t} (3 \sin t - \cos t)}{10}$$

$$(9) f(x)_{rms} = \sqrt{\frac{1}{T} \int_0^T [f(x)]^2 dx}$$

$$a) f(x) = \sqrt{\sin^{-1} x} \text{ over } [0, 1]$$

$$f(x)_{rms} = \sqrt{\frac{1}{1} \int_0^1 (\sqrt{\sin^{-1} x})^2 dx}$$

$$= \sqrt{\int_0^1 \sin^{-1} x dx}$$

$$I = \int \sin^{-1} x dx$$

$$u = \sin^{-1} x \quad dv = dx$$

$$du = \frac{dx}{\sqrt{1-x^2}} \quad v = x$$

$$I = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

$$I_1 = \int \frac{x}{\sqrt{1-x^2}} dx$$

$$\text{let } t = 1-x^2$$

$$dt = -2x dx$$

$$x dx = -\frac{dt}{2}$$

$$I_1 = \int -\frac{dt}{2\sqrt{t}} = -\frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

$$= -\frac{1}{2} (2\sqrt{t}) + C = -\sqrt{1-x^2} + C$$

$$I = \int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1-x^2} + C$$

$$\int_0^1 \sin^{-1} x dx = \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1$$

$$= 1 \sin^{-1} 1 + \sqrt{1-1^2} - (0 \sin^{-1} 0 + \sqrt{1-0^2})$$

$$= \frac{\pi}{2} - 1$$

$$f(x)_{rms} = \sqrt{\frac{\pi}{2} - 1}$$

$$f(x) = \sqrt{e^x \cos x} \text{ over } [0, \frac{\pi}{2}]$$

$$f(x)_{rms} = \sqrt{\frac{2}{\pi} \int_0^{\pi/2} (e^x \cos x)^2 dx}$$

$$= \sqrt{\frac{2}{\pi} \int_0^{\pi/2} e^x \cos x dx}$$

$$I = \int e^x \cos x dx$$

$$u = e^x \quad dv = \cos x dx$$

$$du = e^x dx \quad v = \sin x$$

$$I = e^x \sin x - \int e^x \sin x dx$$

$$I_1 = \int e^x \sin x dx$$

$$u = e^x \quad dv = \sin x dx$$

$$du = e^x dx \quad v = -\cos x$$

$$I_1 = -e^x \cos x + \int e^x \cos x dx$$

$$I_1 = -e^x \cos x + I$$

$$I = e^x \sin x - \int e^x \sin x dx$$

$$I = e^x \sin x + e^x \cos x - I$$

$$2I = e^x \sin x + e^x \cos x$$

$$\int e^x \cos x dx = \frac{1}{2} e^x (\sin x + \cos x) + C$$

$$\int_0^{\pi/2} e^x \cos x = \left[\frac{1}{2} e^x (\sin x + \cos x) \right]_0^{\pi/2}$$

$$= \frac{1}{2} e^{\pi/2} (\sin \frac{\pi}{2} + \cos \frac{\pi}{2}) - \frac{1}{2} e^0 (\sin 0 + \cos 0)$$

$$= \frac{1}{2} e^{\pi/2} - \frac{1}{2} = \frac{1}{2} (e^{\pi/2} - 1)$$

$$f(x)_{rms} = \sqrt{\frac{2}{\pi} \int_0^{\pi/2} e^x \cos x dx}$$

$$= \sqrt{\frac{2}{\pi} \left(\frac{1}{2} (e^{\pi/2} - 1) \right)}$$

$$f(x)_{rms} = \sqrt{\frac{1}{\pi} (e^{\pi/2} - 1)}$$

$$(10) a) \int \frac{\sec^{-1} x}{x \sqrt{x^2 - 1}} dx$$

$$\text{let } u = \sec^{-1} x$$

$$du = \frac{dx}{x \sqrt{x^2 - 1}}$$

$$\int \frac{\sec^{-1} x}{x \sqrt{x^2 - 1}} dx = \int u du = \frac{u^2}{2} + C$$

$$= \boxed{\frac{(\sec^{-1} x)^2}{2} + C}$$

$$b) \int x^3 \sin x^2 dx$$

$$\text{let } u = x^2$$

$$du = 2x dx$$

$$x dx = \frac{du}{2}$$

$$\int x^2 \cdot x \sin x^2 dx = \int \frac{u}{2} \sin u du$$

$$= \frac{1}{2} \int u \sin u du$$

$$\int u \sin u du$$

$$v = u$$

$$dv = du$$

$$dw = \sin u du$$

$$w = -\cos u$$

$$\frac{1}{2} \int u \sin u du = \frac{1}{2} (-u \cos u + \int \cos u du)$$

$$= \frac{1}{2} (\sin u - u \cos u) + C$$

$$\boxed{\int x^3 \sin x^2 dx = \frac{1}{2} (\sin x^2 - x^2 \cos x^2) + C}$$

$$c) \int_0^{\infty} e^{-x} \sin 4x dx$$

$$I = \int e^{-x} \sin 4x dx$$

$$u = \sin 4x$$

$$du = 4 \cos 4x dx$$

$$dv = e^{-x} dx$$

$$v = -e^{-x}$$

$$I = -e^{-x} \sin 4x + 4 \int e^{-x} \cos 4x dx$$

$$I_1 = \int e^{-x} \cos 4x dx$$

$$u = \cos 4x$$

$$du = -4 \sin 4x dx$$

$$dv = e^{-x} dx$$

$$v = -e^{-x}$$

$$I_1 = -e^{-x} \cos 4x - 4 \int e^{-x} \sin 4x dx$$

$$= -e^{-x} \cos 4x - 4I$$

$$I = -e^{-x} \sin 4x + 4(-e^{-x} \cos 4x - 4I)$$

$$I = -e^{-x} \sin 4x - 4e^{-x} \cos 4x + 16I$$

$$17I = -e^{-x} \sin 4x - 4e^{-x} \cos 4x$$

$$I = -\frac{e^{-x}}{17} (\sin 4x + 4 \cos 4x) + C$$

$$\int_0^R e^{-x} \sin 4x dx = J$$

$$J = \left[-\frac{e^{-x}}{17} (\sin 4x + 4 \cos 4x) \right]_0^R$$

$$= -\frac{e^{-R}}{17} (\sin 4R + 4 \cos 4R) + \frac{e^{-0}}{17} (\sin 0 + 4 \cos 0)$$

$$= -\frac{e^{-R}}{17} (\sin 4R + 4 \cos 4R) + \frac{4}{17}$$

$$\lim_{R \rightarrow \infty} J = \lim_{R \rightarrow \infty} \left(-\frac{e^{-R}}{17} (\sin 4R + 4 \cos 4R) + \frac{4}{17} \right)$$

$$= \frac{4}{17}$$

$$\boxed{\int_0^{\infty} e^{-x} \sin 4x dx = \frac{4}{17}}$$

$$d) \int \frac{12x^4 + 190x^2 + 13x - 6}{(2x-1)(x^2+16)} dx = I$$

$$\frac{12x^4 + 190x^2 + 13x - 6}{(2x-1)(x^2+16)} = \frac{A}{2x-1} + \frac{Bx+C}{x^2+16}$$

$$= \frac{A(x^2+16) + (Bx+C)(2x-1)}{(2x-1)(x^2+16)}$$

$$A(x^2+16) + (Bx+C)(2x-1) = Ax^2 + 16A + 2Bx^2 - Bx + 2Cx - C$$

$$= (A+2B)x^2 + (-B+2C)x + 16A - C$$

$$\begin{cases} A+2B = 190 \\ -B+2C = 13 \\ 16A-C = -6 \Rightarrow C = 16A+6 \end{cases}$$

$$-B+2C = 13$$

$$B = 32A - 1$$

$$-B+2(16A+6) = 13$$

$$= 32\left(\frac{192}{65}\right) - 1$$

$$-B+32A+12 = 13$$

$$B = 32A - 1$$

$$B = \frac{6144}{65} - \frac{65}{65}$$

$$A+2B = 190$$

$$B = \frac{6079}{65}$$

$$A+2(32A-1) = 190$$

$$A+64A-2 = 190$$

$$65A = 192$$

$$A = \frac{192}{65}$$

$$C = 16A+6$$

$$= 16\left(\frac{192}{65}\right) + 6$$

$$= \frac{3072}{65} + \frac{390}{65}$$

$$C = \frac{3462}{65}$$

$$\int \frac{A}{2x-1} + \frac{Bx+C}{x^2+16} = \frac{192}{65(2x-1)} + \frac{6079x+3462}{65(x^2+16)}$$

$$\int \frac{12x^4 + 190x^2 + 13x - 6}{(2x-1)(x^2+16)} dx = \int \left(\frac{192}{65(2x-1)} + \frac{6079x+3462}{65(x^2+16)} \right) dx$$

$$\int \frac{192 dx}{65(2x-1)} = \frac{192}{65} \ln|2x-1| + C$$

$$\int \frac{6079x+3462}{65(x^2+16)} dx = \int \frac{6079x dx}{65(x^2+16)} + \int \frac{3462 dx}{65(x^2+16)}$$

$$\frac{6079}{65} \int \frac{x dx}{x^2+16} = \frac{6079}{65} \int \frac{du}{2u} = \frac{6079}{65(2)} \ln|u| + C$$

$$\text{Let } u = x^2+16 \quad \frac{du}{2} = x dx$$

$$= \frac{6079}{65(2)} \ln|x^2+16| + C$$

$$\int \frac{3462 dx}{65(x^2+16)} = \frac{3462}{65(4)} \arctan\left(\frac{x}{4}\right) + C$$

$$I = \frac{192}{65} \ln|2x-1| + \frac{6079}{65(2)} \ln|x^2+16| + \frac{1731}{65(4)} \arctan\left(\frac{x}{4}\right) + C$$

$$\boxed{I = \frac{1}{65} \left[192 \ln|2x-1| + \frac{6079}{2} \ln|x^2+16| + \frac{1731}{2} \arctan\left(\frac{x}{4}\right) \right] + C}$$

$$\int \sec^6 x \tan^8 x dx$$

$$\text{Let } u = \tan x$$

$$du = \sec^2 x dx$$

$$\int \sec^6 x \tan^8 x dx = \int \sec^4 x \tan^8 x \sec^2 x dx$$

$$= \int (\sec^2 x)^2 u^8 du$$

$$= \int (1 + \tan^2 x)^2 u^8 du$$

$$= \int (1 + u^2)^2 u^8 du$$

$$= \int (1 + 2u^2 + u^4) u^8 du$$

$$= \int (u^8 + 2u^{10} + u^{12}) du$$

$$= \frac{u^9}{9} + \frac{2u^{11}}{11} + \frac{u^{13}}{13} + C$$

$$\int \sec^6 x \tan^8 x dx = \frac{\tan^{13} x}{13} + \frac{2 \tan^{11} x}{11} + \frac{\tan^9 x}{9} + C$$

$$f) \int \frac{\cos^4 x}{\sin x} dx$$

$$\frac{1}{\sin x} = \frac{\sin x}{\sin^2 x}$$

$$\int \frac{\cos^4 x \cdot \sin x dx}{\sin^2 x}$$

$$\text{Let } u = \cos x$$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\int \frac{\cos^4 x \sin x dx}{\sin^2 x} = \int \frac{\cos^4 x \sin x dx}{1 - \cos^2 x}$$

$$= - \int \frac{u^4 du}{1 - u^2} = \int \frac{u^4 du}{u^2 - 1}$$

$$\frac{u^4}{u^2 - 1} = \frac{u^2 + 1}{u^2 - 1} + \frac{1}{u^2 - 1}$$

$$\frac{u^4}{u^2 - 1} = u^2 + 1 + \frac{1}{u^2 - 1}$$

$$\int \frac{u^4 du}{u^2 - 1} = \int \left(u^2 + 1 + \frac{1}{u^2 - 1} \right) du$$

$$= \frac{u^3}{3} + u + \frac{1}{2} \ln \left| \frac{u-1}{u+1} \right| + C$$

$$\int \frac{du}{u^2 - 1} = \frac{1}{2} \ln \left| \frac{u-1}{u+1} \right| + C$$

$$\int \frac{\cos^4 x}{\sin x} dx = \frac{\cos^3 x}{3} + \cos x + \frac{1}{2} \ln \left| \frac{\cos x - 1}{\cos x + 1} \right| + C$$

$$g) \int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$$

$$\text{Let } x = \sin^2 \theta$$

$$dx = 2 \sin \theta \cos \theta d\theta$$

$$\int \frac{1}{\sqrt{x(1-x)}} dx = \int \frac{2 \sin \theta \cos \theta d\theta}{\sqrt{\sin^2 \theta (1 - \sin^2 \theta)}}$$

$$= \int \frac{2 \sin \theta \cos \theta d\theta}{\sqrt{\sin^2 \theta \cos^2 \theta}}$$

$$= \int \frac{2 \sin \theta \cos \theta d\theta}{\sin \theta \cos \theta}$$

$$= \int 2 d\theta = 2\theta + C$$

$$\text{If } x = \sin^2 \theta \Rightarrow \sqrt{x} = \sin \theta$$

$$\Rightarrow \theta = \arcsin \sqrt{x}$$

At

$$\text{At } x=0, \theta = \arcsin \sqrt{0} = 0$$

$$\text{At } x=1, \theta = \arcsin \sqrt{1} = \frac{\pi}{2}$$

$$\int_0^{\pi/2} 2 d\theta = \left[2\theta \right]_0^{\pi/2} = 2\left(\frac{\pi}{2}\right) - 2(0)$$

$$= \pi$$

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx = \pi$$

⑪ $y = x^3$ and $y = |x|$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Intersection

If $|x| = x$

$$y = x^3, y = x$$

$$x^3 = x$$

$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x = 0, x^2 - 1 = 0$$

$$x = \pm 1$$

$$x = 0, x = -1, x = 1$$

If $|x| = -x$

$$y = x^3, y = -x$$

$$x^3 = -x$$

$$x^3 + x = 0$$

$$x(x^2 + 1) = 0$$

$$x = 0, x^2 + 1 = 0$$

$$x^2 \neq -1$$

No enclosed area between $y = x^3$ and $y = -x$

$$\begin{aligned} \text{Area} &= \int_{-1}^0 (x^3 - x) dx + \int_0^1 (x - x^3) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 \\ &= 0 - \left[\frac{(-1)^4}{4} - \frac{(-1)^2}{2} \right] + \frac{1^2}{2} - \frac{1^4}{4} - 0 = -\frac{1}{4} + \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = -\frac{2}{4} + 1 = \frac{1}{2} \end{aligned}$$

$$\boxed{\text{Area} = \frac{1}{2}}$$

⑫ $f(x) = 12 + x^2$

$$g(x) = x$$

$$\text{Volume} = \pi \int_0^2 (12 + x^2 - x)^2 dx$$

$$= \pi \int_0^2 (144 + 24x^2 + x^4 - x^2) dx$$

$$= \pi \int_0^2 (144 + 23x^2 + x^4) dx$$

$$= \pi \left[144x + \frac{23}{3}x^3 + \frac{x^5}{5} \right]_0^2$$

$$= \pi \left[\left(144(2) + \frac{23}{3}(2)^3 + \frac{2^5}{5} \right) - \left(144(0) + \frac{23}{3}(0)^3 + \frac{0^5}{5} \right) \right]$$

$$= \pi \left(288 + \frac{23}{3}(8) + \frac{32}{5} \right)$$

$$= \pi \left(\frac{4320 + 920 + 96}{15} \right)$$

$$\boxed{\text{Volume} = \frac{5336}{15} \pi}$$